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root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
%cuts_list = cuts2fit200.cutsS_list();
%cuts_list = {cuts2fit200.cut200_200sel};
%cuts_list = {cuts2fit200.cut110_sel};
%cuts_list = {cut(cuts2fit200.cut200_sel, [0, 0.02, 1], [])};
cuts_list = cuts2fit200.cuts_dir001FFfit(1);
mi = maps_instrument(200, 600, 'S');
sample = IX_sample(true, [1, 0, 0], [0, 1, 0], 'cuboid', [0.04, 0.03, 0.02]);
for i = 1: numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
end

dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
%cut_en = 30:10:120;
%cut_en = 40:10:150;
%cut_en = 145;
cut_en = 30:10:110; % cuts2fit200.cuts_dir001FFfit{1}

%cl = {cuts_list{3}};
%cl = {w2_test_cut};
if isfield(cuts2fit200, 'bg_par_010off000')
    if numel(cuts_list) == 3
        bg_par =
{cuts2fit200.bg_par_010off000.p, cuts2fit200.bg_par_010off100.p, cuts2fit200.bg_par
_010off200.p};
    else
        %bg_par = {cuts2fit200.bg_par_010off100.p};
        bg_par = {cuts2fit200.bg_par_010off200.p};
    end
    cuts_list{end+1} = bg_par;
end

[fit_res_200fix_bg, res_name] = fit_multicuts_along_direction(...

cuts_list, 'EnFit_Ei200ref001off200_2D_constFFCut', dir_name, cut_en, dE_step, half_dE
,true);

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*****
*** En = 30±10
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Beginning fit (max 20 iterations)-----

Starting point

Total time = 3.8839s Reduced Chi^2 = 111.3654

Free parameter values:

10 1 30 4.063

Iteration = 1

Total time = 15.5268s Reduced Chi^2 = 8.8533 Levenberg-Marquardt = 0.1

Free parameter values:

7.412 0.8343 23.12 1.843

Iteration = 2

Total time = 26.3885s Reduced Chi^2 = 7.4191 Levenberg-Marquardt = 0.01

Free parameter values:

7.493 0.9323 24.25 1.599

Iteration = 3

Total time = 37.2077s Reduced Chi^2 = 7.3806 Levenberg-Marquardt = 0.001

Free parameter values:

6.812 0.905 24.19 1.624

Iteration = 4

Total time = 47.9865s Reduced Chi^2 = 7.348 Levenberg-Marquardt = 0.0001

Free parameter values:

6.757 0.9028 24.24 1.628

Iteration = 5

Total time = 58.7857s Reduced Chi^2 = 7.3586 Levenberg-Marquardt = 1e-05

Free parameter values:

6.712 0.9023 24.24 1.629

Total time = 61.4996s Reduced Chi^2 = 7.3815 Levenberg-Marquardt = 0.0001

Free parameter values:

6.712 0.9023 24.24 1.629

Total time = 64.2085s Reduced Chi^2 = 7.3747 Levenberg-Marquardt = 0.001

Free parameter values:

6.712 0.9023 24.24 1.629

Total time = 66.9209s Reduced Chi^2 = 7.3682 Levenberg-Marquardt = 0.1

Free parameter values:

6.717 0.9026 24.24 1.629

Total time = 69.6552s Reduced Chi^2 = 7.3501 Levenberg-Marquardt = 10

Free parameter values:

6.745 0.9029 24.24 1.628

Total time = 72.3645s Reduced Chi^2 = 7.3846 Levenberg-Marquardt = 1000

Free parameter values:

6.755 0.9028 24.24 1.628

*** No improvement in chi-squared over previous iteration ***

Total time = 72.372s Reduced Chi^2 = 7.348

Free parameter values:

6.757 0.9028 24.24 1.628

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	6.757	+/-	0.1755
4	0.9028	+/-	0.02406
5	0		

6	24.24	+/-	0.2266
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	1.628	+/-	0.06249
2	-0.04831		
3	0		

Covariance matrix for free parameters:

1.0000	0.1217	-0.3124	-0.6849
0.1217	1.0000	-0.0354	-0.7689
-0.3124	-0.0354	1.0000	-0.0081
-0.6849	-0.7689	-0.0081	1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 278883 pixels -- processing data... -----> included 251146 pixels

Number of points in input object: 278883
 Fraction of object selected: 100.00% (= 278883 points) ; Access speed: 1.97GB/sec
 Fraction of object retained: 90.05% (= 251146 points)

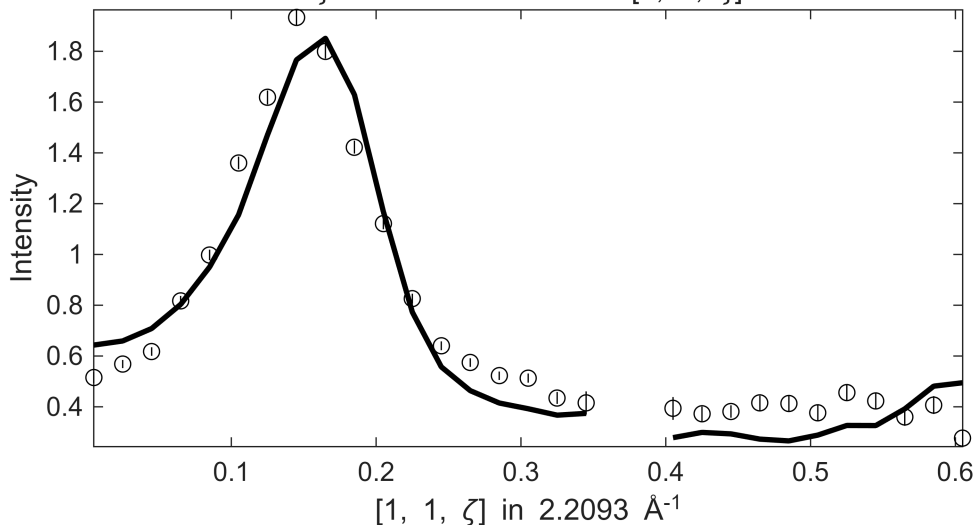
**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 40.5%; Transf. time frac.: 51.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 278883 pixels -- processing data... -----> retained 251146 pixels

Number of points in input object: 278883
 Fraction of object selected: 100.00% (= 278883 points) ; Access speed: 1.92GB/sec
 Fraction of object retained: 90.05% (= 251146 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 20.0%; Transf. time frac.: 54.7%; Resulting pix preparation: 23.5%

Fe_ei200_align_sym3.sqw
 S= 0.9±0.024; J0= 24±0.23; gamma=6.75665±0.175542
 -0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 20 ≤ E ≤ 40
 ζ=-0.005:0.02:0.615 in [1, 1, ζ]



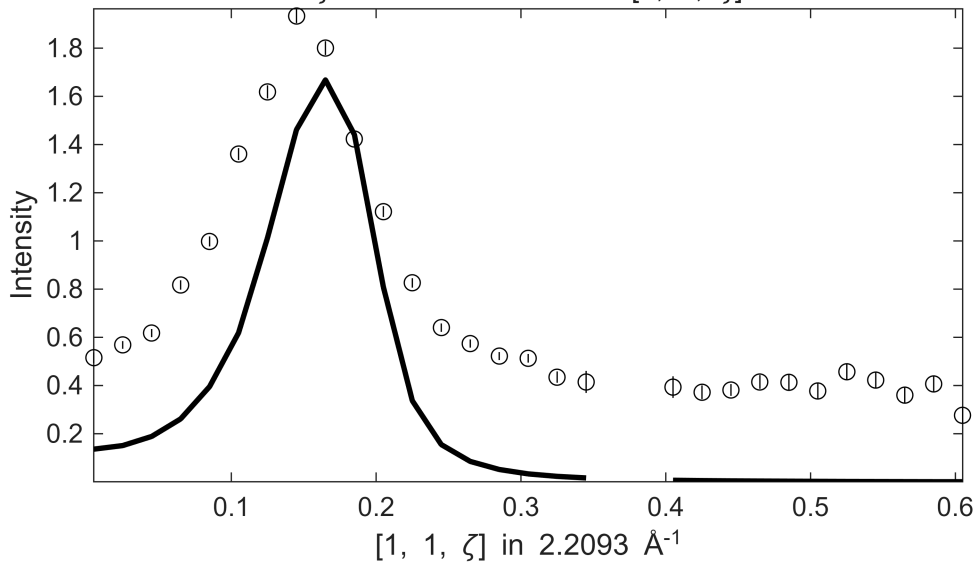
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 278883 pixels -- processing data... -----> retained 251146 pixels

Number of points in input object: 278883

Fraction of object selected: 100.00% (= 278883 points) ; Access speed: 2.01GB/sec
 Fraction of object retained: 90.05% (= 251146 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 21.6%; Transf. time frac.: 52.0%; Resulting pix preparation: 26.4%

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $20 \leq E \leq 40$
 $\zeta = -0.005:0.02:0.615$ in $[1, 1, \zeta]$



**** En = 40±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.29326

Beginning fit (max 20 iterations)-----

Starting point

 Total time = 3.3737s Reduced Chi^2 = 66.6766

Free parameter values:

6.757	0.9028	24.24	3.282
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 Iteration = 1

 Total time = 17.351s Reduced Chi^2 = 4.0089 Levenberg-Marquardt = 0.1

Free parameter values:

8.059	0.7296	24.27	1.607
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 Iteration = 2

 Total time = 31.5654s Reduced Chi^2 = 3.8343 Levenberg-Marquardt = 0.01

Free parameter values:

9.386	0.7729	24.31	1.48
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 Iteration = 3

 Total time = 45.8038s Reduced Chi^2 = 3.8163 Levenberg-Marquardt = 0.001

Free parameter values:

9.458	0.7758	24.24	1.477
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 Iteration = 4

 Total time = 60.1144s Reduced Chi^2 = 3.8086 Levenberg-Marquardt = 0.0001

Free parameter values:

9.496	0.7763	24.22	1.476
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 Iteration = 5

Total time = 74.3912s Reduced Chi^2 = 3.8067 Levenberg-Marquardt = 1e-05

Free parameter values:

9.461 0.7763 24.22 1.476

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1 1
2 8
3 9.461 +/- 0.64
4 0.7763 +/- 0.0285
5 0
6 24.22 +/- 0.1978
7 0
8 0
9 0
10 0

Background parameter values:

1 1.476 +/- 0.04225
2 -0.04223
3 0

Covariance matrix for free parameters:

1.0000 0.8014 0.3988 -0.5694
0.8014 1.0000 0.3862 -0.7174
0.3988 0.3862 1.0000 -0.2337
-0.5694 -0.7174 -0.2337 1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 364485 pixels -- processing data... -----> included 332114 pixels

Number of points in input object: 364485

Fraction of object selected: 100.00% (= 364485 points) ; Access speed: 1.71GB/sec
Fraction of object retained: 91.12% (= 332114 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 44.2%; Transf. time frac.: 53.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 364481 pixels -- processing data... -----> retained 332110 pixels

Number of points in input object: 364481

Fraction of object selected: 100.00% (= 364481 points) ; Access speed: 1.86GB/sec
Fraction of object retained: 91.12% (= 332110 points)

**** Total time in cut(sqw) : 0.0sec Including:

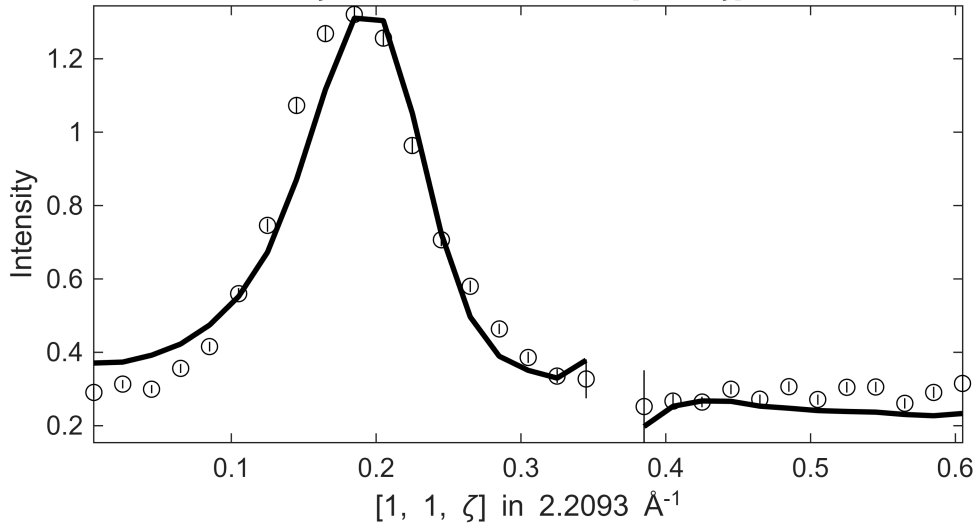
Data access time fraction: 30.5%; Transf. time frac.: 49.4%; Resulting pix preparation: 20.1%

Fe_ei200_align_sym3.sqw

S=0.78±0.029; J0= 24± 0.2; gamma=9.46078±0.63997

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 30 ≤ E ≤ 50

ζ=-0.005:0.02:0.615 in [1, 1, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 364485 pixels -- processing data... -----> retained 332114 pixels

Number of points in input object: 364485

Fraction of object selected: 100.00% (= 364485 points) ; Access speed: 1.96GB/sec

Fraction of object retained: 91.12% (= 332114 points)

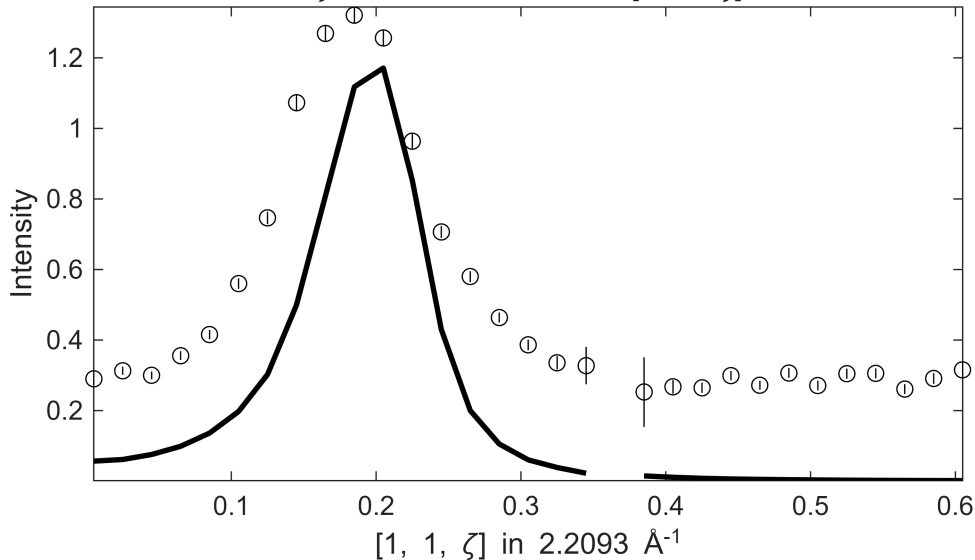
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 25.7%; Transf. time frac.: 54.5%; Resulting pix preparation: 19.8%

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 30 ≤ E ≤ 50

ζ=-0.005:0.02:0.615 in [1, 1, ζ]



**** En = 50±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.29326

Beginning fit (max 20 iterations)-----

Starting point

Total time = 5.0288s Reduced Chi^2 = 52.0114

Free parameter values:

	9.461	0.7763	24.22	1.803

Iteration = 1				

Total time = 21.567s	Reduced Chi^2 = 3.0591		Levenberg-Marquardt = 0.1	
Free parameter values:				
12.86	0.7551	24.68	0.8324	

Iteration = 2				

Total time = 38.0187s	Reduced Chi^2 = 2.8674		Levenberg-Marquardt = 0.01	
Free parameter values:				
15.31	0.8278	25	0.7541	

Iteration = 3				

Total time = 54.491s	Reduced Chi^2 = 2.8656		Levenberg-Marquardt = 0.001	
Free parameter values:				
15.3	0.8289	24.95	0.7547	

Fit converged:				
Parameter values (free parameters with error estimates):				

Foreground parameter values:				

1	1			
2	8			
3	15.3	+/-	0.826	
4	0.8289	+/-	0.02714	
5	0			
6	24.95	+/-	0.2047	
7	0			
8	0			
9	0			
10	0			
Background parameter values:				

1	0.7547	+/-	0.01907	
2	-0.02868			
3	0			
Covariance matrix for free parameters:				

1.0000	0.8172	0.4506	-0.5149	
0.8172	1.0000	0.4190	-0.6650	
0.4506	0.4190	1.0000	-0.2530	
-0.5149	-0.6650	-0.2530	1.0000	
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested				
*** Step 1 of 1; Read data for 445825 pixels -- processing data... -----> included 404672 pixels				

Number of points in input object: 445825				
Fraction of object selected: 100.00% (= 445825 points) ; Access speed: 1.18GB/sec				
Fraction of object retained: 90.77% (= 404672 points)				
**** Total time in cut(sqw) : 0.1sec Including:				
Data access time fraction: 34.4%; Transf. time frac.: 62.4%; Resulting pix preparation: 0.0%				

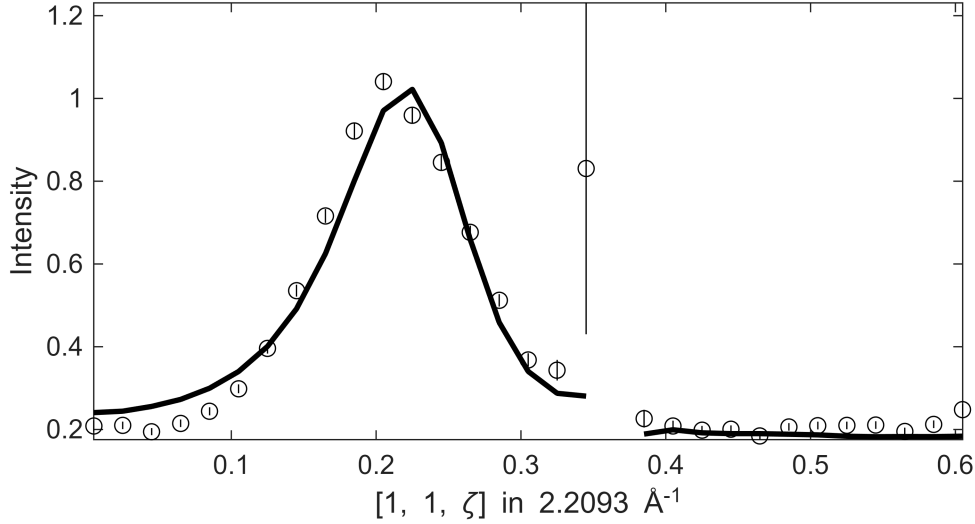
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept				
*** Step 1 of 1; Read data for 445821 pixels -- processing data... -----> retained 404668 pixels				

Number of points in input object: 445821				
Fraction of object selected: 100.00% (= 445821 points) ; Access speed: 1.76GB/sec				

Fraction of object retained: 90.77% (= 404668 points)

*** Total time in cut(sqw) : 0.1sec Including:
Data access time fraction: 16.5%; Transf. time frac.: 39.8%; Resulting pix preparation: 42.4%

Fe_ei200_align_sym3.sqw
S=0.83±0.027; J0= 25± 0.2; gamma=15.3029±0.825977
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 40 ≤ E ≤ 60
ζ=-0.005:0.02:0.615 in [1, 1, ζ]

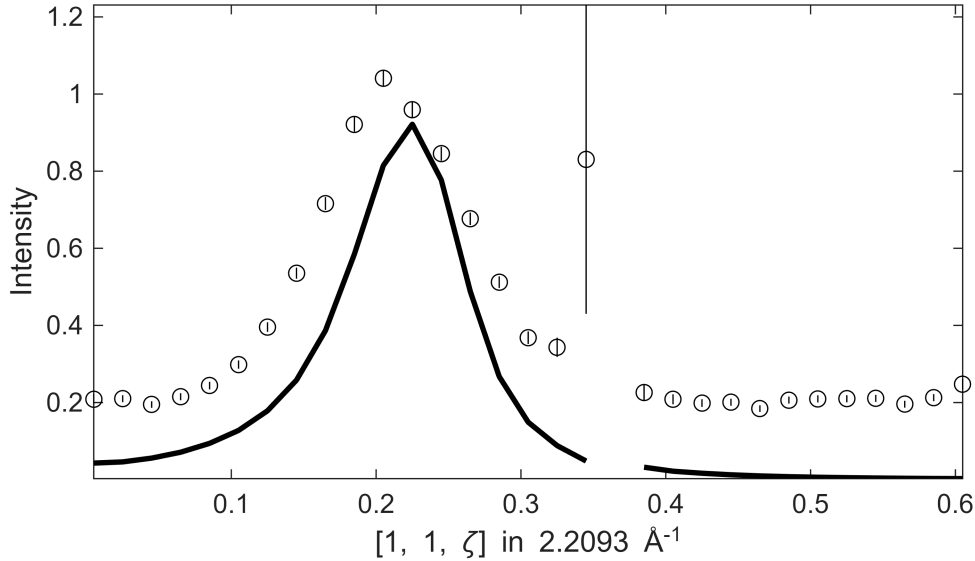


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 445825 pixels -- processing data... -----> retained 404672 pixels

Number of points in input object: 445825
Fraction of object selected: 100.00% (= 445825 points) ; Access speed: 2.09GB/sec
Fraction of object retained: 90.77% (= 404672 points)

*** Total time in cut(sqw) : 0.1sec Including:
Data access time fraction: 27.4%; Transf. time frac.: 52.2%; Resulting pix preparation: 20.4%

Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 40 ≤ E ≤ 60
ζ=-0.005:0.02:0.615 in [1, 1, ζ]



*** En = 60±10

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.58651

Beginning fit (max 20 iterations)-----

Starting point

Total time = 4.7107s Reduced Chi^2 = 51.588

Free parameter values:

15.3 0.8289 24.95 1.414

Iteration = 1

Total time = 23.3268s Reduced Chi^2 = 2.5239 Levenberg-Marquardt = 0.1

Free parameter values:

18.09 0.7505 25.69 0.6396

Iteration = 2

Total time = 42.0259s Reduced Chi^2 = 2.3767 Levenberg-Marquardt = 0.01

Free parameter values:

20.33 0.8061 26.15 0.5821

Iteration = 3

Total time = 60.375s Reduced Chi^2 = 2.3778 Levenberg-Marquardt = 0.001

Free parameter values:

20.19 0.8046 26.14 0.5829

Total time = 64.9862s Reduced Chi^2 = 2.3794 Levenberg-Marquardt = 0.01

Free parameter values:

20.19 0.8047 26.14 0.5829

Total time = 69.6176s Reduced Chi^2 = 2.3772 Levenberg-Marquardt = 0.1

Free parameter values:

20.21 0.8052 26.14 0.5827

Total time = 74.1993s Reduced Chi^2 = 2.3798 Levenberg-Marquardt = 10

Free parameter values:

20.3 0.8062 26.15 0.5822

Total time = 78.8803s Reduced Chi^2 = 2.3759 Levenberg-Marquardt = 1000

Free parameter values:

20.32 0.8061 26.15 0.5821

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1 1
2 8
3 20.32 +/- 1.081
4 0.8061 +/- 0.0279
5 0
6 26.15 +/- 0.2228
7 0
8 0
9 0
10 0

Background parameter values:

1 0.5821 +/- 0.01472
2 -0.02443
3 0

Covariance matrix for free parameters:

1.0000 0.8517 0.3516 -0.5641
0.8517 1.0000 0.2772 -0.6930

0.3516	0.2772	1.0000	-0.2042
-0.5641	-0.6930	-0.2042	1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 514917 pixels -- processing data... -----> included 467110 pixels

 Number of points in input object: 514917
 Fraction of object selected: 100.00% (= 514917 points) ; Access speed: 1.76GB/sec
 Fraction of object retained: 90.72% (= 467110 points)

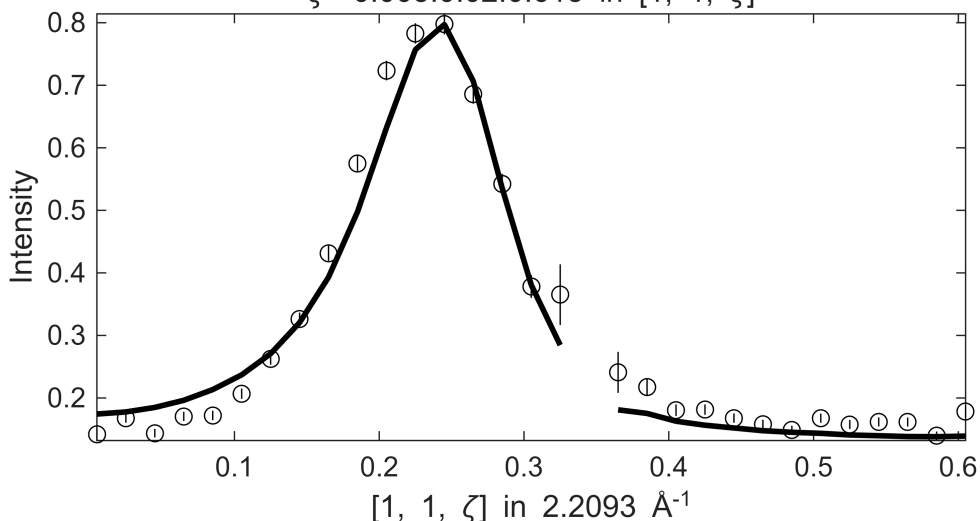
**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 53.7%; Transf. time frac.: 44.8%; Resulting pix preparation: 0.0%

 *** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 514896 pixels -- processing data... -----> retained 467089 pixels

 Number of points in input object: 514896
 Fraction of object selected: 100.00% (= 514896 points) ; Access speed: 2.17GB/sec
 Fraction of object retained: 90.72% (= 467089 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 31.6%; Transf. time frac.: 47.3%; Resulting pix preparation: 21.1%

 Fe_ei200_align_sym3.sqw
 S=0.81±0.028; J0= 26±0.22; gamma=20.3243±1.08108
 -0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 50 ≤ E ≤ 70 eV
 ζ=-0.005:0.02:0.615 in [1, 1, ζ]



 *** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 514917 pixels -- processing data... -----> retained 467110 pixels

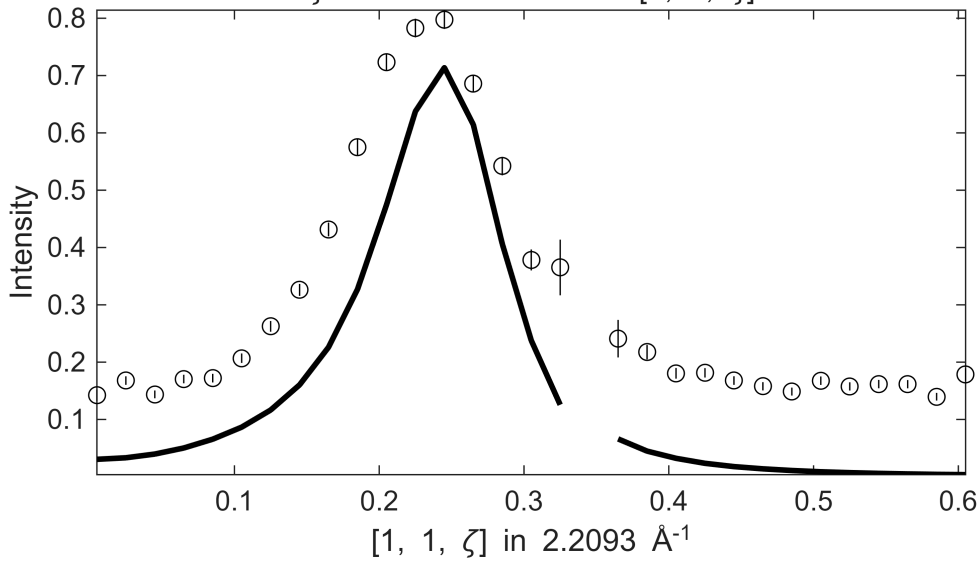
 Number of points in input object: 514917
 Fraction of object selected: 100.00% (= 514917 points) ; Access speed: 2.10GB/sec
 Fraction of object retained: 90.72% (= 467110 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 26.5%; Transf. time frac.: 56.1%; Resulting pix preparation: 17.4%

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $50 \leq E \leq 100$ eV

$\zeta = -0.005:0.02:0.615$ in $[1, 1, \zeta]$



**** En = 70±10

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.87977

Beginning fit (max 20 iterations)-----

Starting point

Total time = 5.2994s Reduced Chi^2 = 43.1735

Free parameter values:

20.32 0.8061 26.15 1.819

Iteration = 1

Total time = 26.3739s Reduced Chi^2 = 2.2715 Levenberg-Marquardt = 0.1

Free parameter values:

22.65 0.7386 26.72 0.8381

Iteration = 2

Total time = 47.4335s Reduced Chi^2 = 2.1405 Levenberg-Marquardt = 0.01

Free parameter values:

25.52 0.8073 26.94 0.7545

Iteration = 3

Total time = 68.4442s Reduced Chi^2 = 2.1374 Levenberg-Marquardt = 0.001

Free parameter values:

25.9 0.8201 26.83 0.7488

Iteration = 4

Total time = 89.5898s Reduced Chi^2 = 2.1372 Levenberg-Marquardt = 0.0001

Free parameter values:

26.14 0.8258 26.8 0.746

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1

1

2	8		
3	26.14	+/-	1.367
4	0.8258	+/-	0.03106
5	0		
6	26.8	+/-	0.2674
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.746	+/-	0.02116
2	-0.02824		
3	0		

Covariance matrix for free parameters:

1.0000	0.8485	0.2998	-0.6398
0.8485	1.0000	0.0828	-0.7623
0.2998	0.0828	1.0000	-0.0899
-0.6398	-0.7623	-0.0899	1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 600900 pixels -- processing data... -----> included 545834 pixels

Number of points in input object: 600900
 Fraction of object selected: 100.00% (= 600900 points) ; Access speed: 2.14GB/sec
 Fraction of object retained: 90.84% (= 545834 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 54.1%; Transf. time frac.: 43.8%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 600875 pixels -- processing data... -----> retained 545809 pixels

Number of points in input object: 600875
 Fraction of object selected: 100.00% (= 600875 points) ; Access speed: 2.09GB/sec
 Fraction of object retained: 90.84% (= 545809 points)

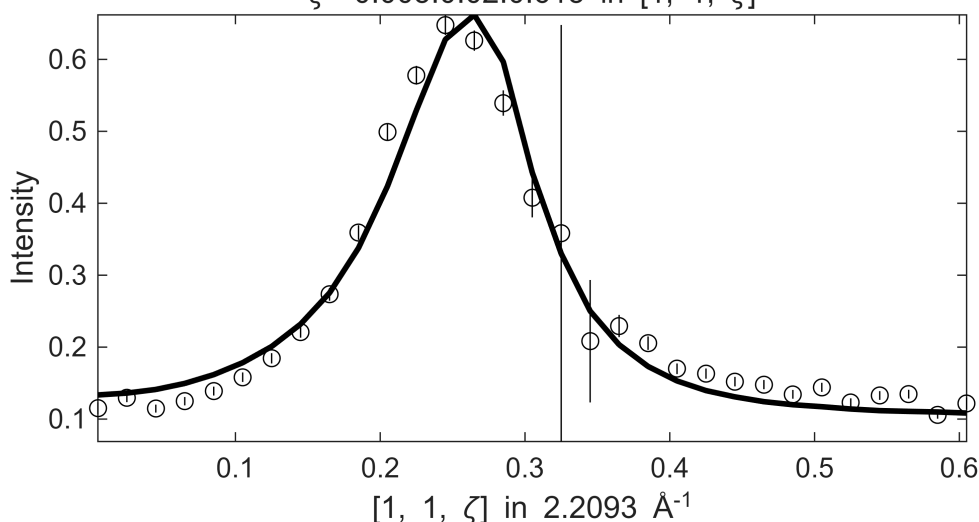
**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 26.0%; Transf. time frac.: 47.3%; Resulting pix preparation: 26.6%

Fe_ei200_align_sym3.sqw

S=0.83±0.031; J0= 27±0.27; gamma=26.1413±1.36711

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 60 ≤ E ≤ 100

ζ=-0.005:0.02:0.615 in [1, 1, ζ]



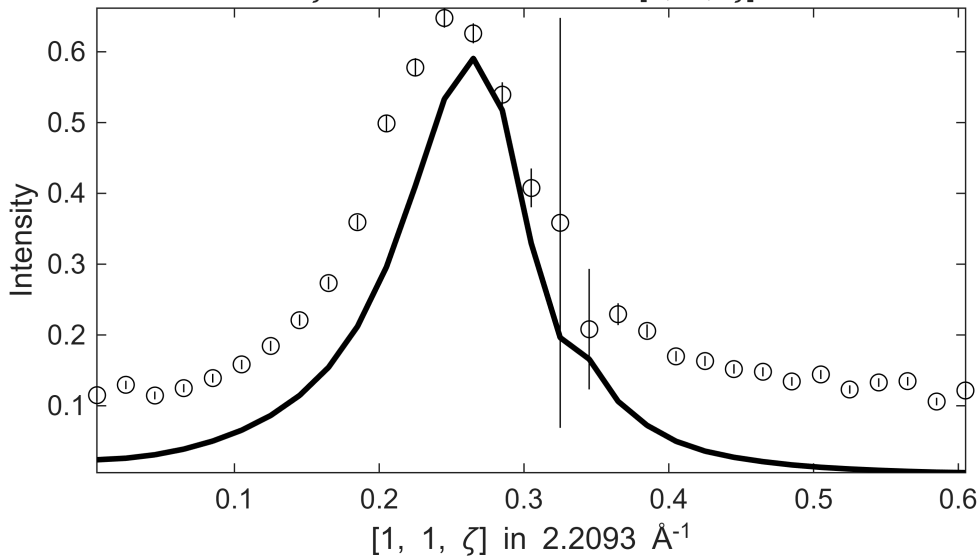
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*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 600900 pixels -- processing data... -----> retained 545834 pixels
-----
Number of points in input object: 600900
Fraction of object selected: 100.00% (= 600900 points) ; Access speed: 2.14GB/sec
Fraction of object retained: 90.84% (= 545834 points)

**** Total time in cut(sqw) : 0.1sec Including:
Data access time fraction: 31.4%; Transf. time frac.: 47.9%; Resulting pix preparation: 20.7%
-----

```

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $60 \leq E \leq 100$ eV
 $\zeta = -0.005:0.02:0.615$ in $[1, 1, \zeta]$



```

*****
**** En = 80±10
*****
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.58651
-----

```

Beginning fit (max 20 iterations)-----
Starting point

```

-----
Total time = 6.2292s    Reduced Chi^2 = 37.6863
Free parameter values:
    26.14      0.8258      26.8      2.332
-----

```

Iteration = 1

```

-----
Total time = 33.2224s    Reduced Chi^2 = 2.0116    Levenberg-Marquardt = 0.1
Free parameter values:
    27.63      0.772      27.08      1.019
-----

```

Iteration = 2

```

-----
Total time = 58.9337s    Reduced Chi^2 = 1.8784    Levenberg-Marquardt = 0.01
Free parameter values:
    30.88      0.8442      27.27      0.8921
-----

```

Iteration = 3

```

-----
Total time = 84.5725s    Reduced Chi^2 = 1.8745    Levenberg-Marquardt = 0.001
Free parameter values:
    31.1      0.8535      27.19      0.8846
-----

```

Iteration = 4

```

-----
Total time = 109.9627s    Reduced Chi^2 = 1.8735    Levenberg-Marquardt = 0.0001
Free parameter values:

```

31.23	0.8552	27.2	0.8828
-------	--------	------	--------

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	31.23	+/-	1.496
4	0.8552	+/-	0.02523
5	0		
6	27.2	+/-	0.2701
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.8828	+/-	0.02863
2	-0.03165		
3	0		

Covariance matrix for free parameters:

1.0000	0.7251	0.6655	-0.5859
0.7251	1.0000	0.3285	-0.7705
0.6655	0.3285	1.0000	-0.2498
-0.5859	-0.7705	-0.2498	1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 718491 pixels -- processing data... -----> included 649651 pixels

Number of points in input object: 718491

Fraction of object selected:	100.00% (= 718491 points)	Access speed:	2.09GB/sec
Fraction of object retained:	90.42% (= 649651 points)		

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction:	47.2%; Transf. time frac.: 51.3%; Resulting pix preparation:	0.0%
----------------------------	--	------

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 718457 pixels -- processing data... -----> retained 649647 pixels

Number of points in input object: 718457

Fraction of object selected:	100.00% (= 718457 points)	Access speed:	2.09GB/sec
Fraction of object retained:	90.42% (= 649647 points)		

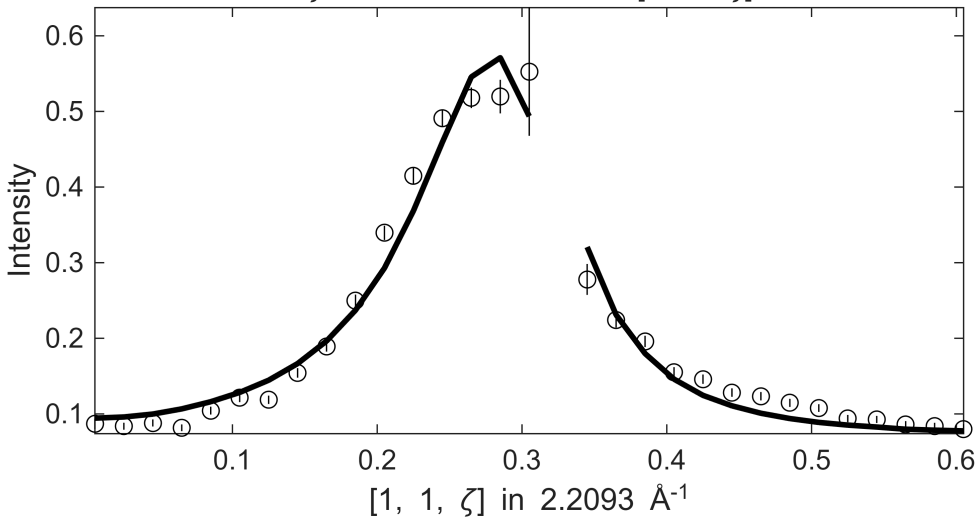
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction:	26.9%; Transf. time frac.: 53.5%; Resulting pix preparation:	19.6%
----------------------------	--	-------

Fe_ei200_align_sym3.sqw

S=0.86±0.025; J0= 27±0.27; gamma=31.2255±1.49637

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 70 ≤ E ≤ 90
ζ=-0.005:0.02:0.615 in [1, 1, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 718491 pixels -- processing data... -----> retained 649651 pixels

Number of points in input object: 718491

Fraction of object selected: 100.00% (= 718491 points) ; Access speed: 1.74GB/sec

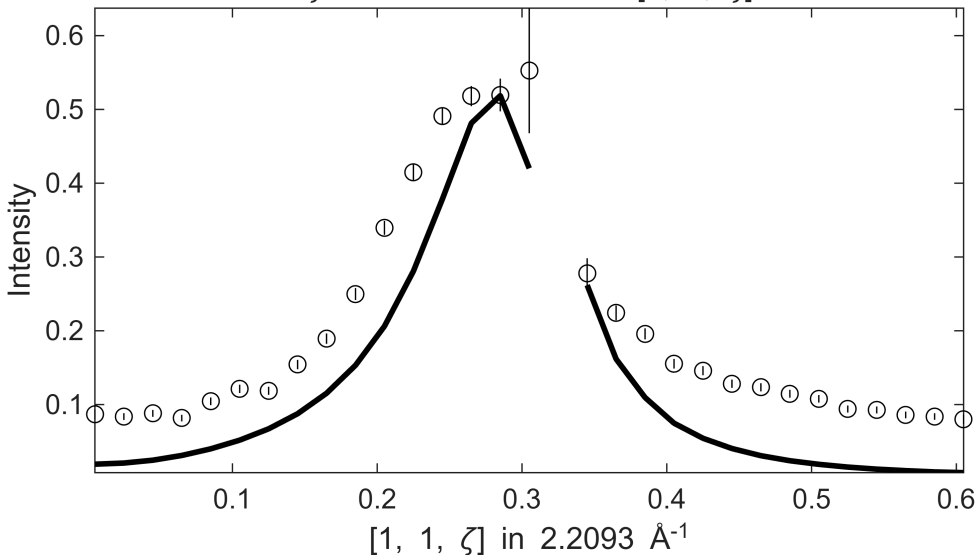
Fraction of object retained: 90.42% (= 649651 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 28.7%; Transf. time frac.: 54.0%; Resulting pix preparation: 17.3%

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 70 ≤ E ≤ 90
ζ=-0.005:0.02:0.615 in [1, 1, ζ]



**** En = 90±10

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.87977

Beginning fit (max 20 iterations)-----

Starting point

Total time = 8.4001s Reduced Chi^2 = 44.0087

Free parameter values:

31.23	0.8552	27.2	0.9006
-------	--------	------	--------

Iteration = 1

Total time = 42.944s	Reduced Chi^2 = 2.3167	Levenberg-Marquardt = 0.1
Free parameter values:		
35.06	0.7917	28.28
		0.348

Iteration = 2

Total time = 76.6964s	Reduced Chi^2 = 2.1704	Levenberg-Marquardt = 0.01
Free parameter values:		
40.05	0.8667	28.92
		0.2908

Iteration = 3

Total time = 109.9674s	Reduced Chi^2 = 2.1655	Levenberg-Marquardt = 0.001
Free parameter values:		
40.23	0.8723	28.84
		0.2891

Iteration = 4

Total time = 143.4554s	Reduced Chi^2 = 2.1684	Levenberg-Marquardt = 0.0001
Free parameter values:		
40.67	0.8764	28.88
		0.2874
Total time = 151.8441s	Reduced Chi^2 = 2.1657	Levenberg-Marquardt = 0.001
Free parameter values:		
40.67	0.8763	28.88
		0.2874
Total time = 160.1665s	Reduced Chi^2 = 2.1662	Levenberg-Marquardt = 0.01
Free parameter values:		
40.65	0.8761	28.88
		0.2875
Total time = 168.549s	Reduced Chi^2 = 2.1668	Levenberg-Marquardt = 1
Free parameter values:		
40.39	0.8733	28.85
		0.2886
Total time = 176.8823s	Reduced Chi^2 = 2.1665	Levenberg-Marquardt = 100
Free parameter values:		
40.25	0.8724	28.84
		0.289
Total time = 185.2231s	Reduced Chi^2 = 2.1687	Levenberg-Marquardt = 10000
Free parameter values:		
40.24	0.8723	28.84
		0.2891

*** No improvement in chi-squared over previous iteration ***

Total time = 185.2324s	Reduced Chi^2 = 2.1655
Free parameter values:	
40.23	0.8723
	28.84
	0.2891

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	40.23	+/-	1.013
4	0.8723	+/-	0.02084
5	0		
6	28.84	+/-	0.1747
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.2891	+/-	0.0128
2	-0.02032		

Covariance matrix for free parameters:

```
-----
1.0000    0.4836    0.2618   -0.8636
0.4836    1.0000    0.2140   -0.8436
0.2618    0.2140    1.0000   -0.4000
-0.8636   -0.8436   -0.4000    1.0000
-----
```

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 989435 pixels -- processing data... -----> included 882648 pixels

Number of points in input object: 989435

Fraction of object selected: 100.00% (= 989435 points) ; Access speed: 1.71GB/sec

Fraction of object retained: 89.21% (= 882648 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 62.3%; Transf. time frac.: 36.7%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 989236 pixels -- processing data... -----> retained 882530 pixels

Number of points in input object: 989236

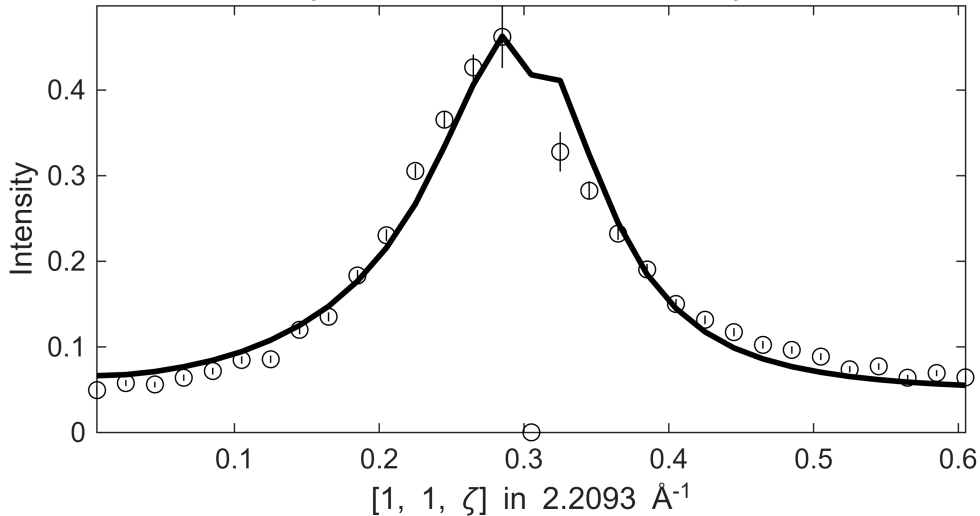
Fraction of object selected: 100.00% (= 989236 points) ; Access speed: 2.13GB/sec

Fraction of object retained: 89.21% (= 882530 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 26.7%; Transf. time frac.: 44.2%; Resulting pix preparation: 29.1%

Fe_ei200_align_sym3.sqw
 $S=0.87\pm0.021$; $J_0=29\pm0.17$; $\gamma=40.2346\pm1.01296$
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $80 \leq E \leq 1$
 $\zeta=-0.005:0.02:0.615$ in $[1, 1, \zeta]$



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 989435 pixels -- processing data... -----> retained 882648 pixels

Number of points in input object: 989435

Fraction of object selected: 100.00% (= 989435 points) ; Access speed: 2.19GB/sec

Fraction of object retained: 89.21% (= 882648 points)

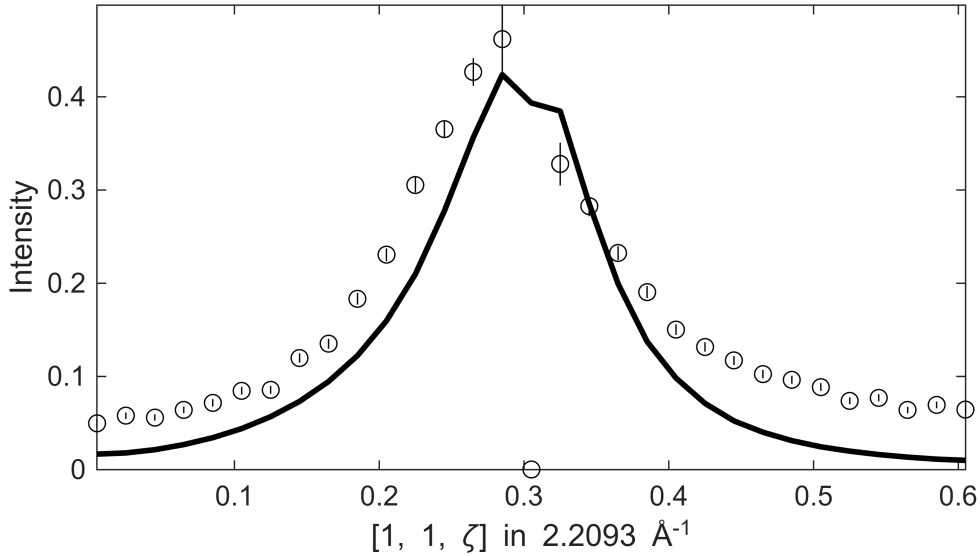
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 28.8%; Transf. time frac.: 50.7%; Resulting pix preparation: 20.4%

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $80 \leq E \leq 1$

$\zeta = -0.005:0.02:0.615$ in $[1, 1, \zeta]$



**** En = 100±10

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.4663

Beginning fit (max 20 iterations)-----

Starting point

Total time = 10.4462s Reduced Chi^2 = 54.3887

Free parameter values:

40.23 0.8723 28.84 0.2905

Iteration = 1

Total time = 52.0002s Reduced Chi^2 = 2.3786 Levenberg-Marquardt = 0.1

Free parameter values:

44.64 0.7871 29.53 0.09519

Iteration = 2

Total time = 93.2635s Reduced Chi^2 = 2.1499 Levenberg-Marquardt = 0.01

Free parameter values:

54.48 0.904 30.47 0.07057

Iteration = 3

Total time = 134.3103s Reduced Chi^2 = 2.1314 Levenberg-Marquardt = 0.001

Free parameter values:

58.39 0.944 30.75 0.06686

Iteration = 4

Total time = 175.3661s Reduced Chi^2 = 2.1282 Levenberg-Marquardt = 0.0001

Free parameter values:

59.91 0.9569 30.85 0.06558

Iteration = 5

Total time = 216.4792s Reduced Chi^2 = 2.1296 Levenberg-Marquardt = 1e-05

Free parameter values:

60.54 0.9619 30.89 0.06504

Total time = 226.9003s Reduced Chi^2 = 2.1293 Levenberg-Marquardt = 0.0001

Free parameter values:

60.54 0.9619 30.89 0.06504

```

Total time = 237.2353s    Reduced Chi^2 = 2.1289    Levenberg-Marquardt = 0.001
Free parameter values:
    60.53    0.9619    30.89    0.06505
Total time = 247.5478s    Reduced Chi^2 = 2.1296    Levenberg-Marquardt = 0.1
Free parameter values:
    60.31    0.9595    30.86    0.06528
Total time = 257.908s    Reduced Chi^2 = 2.1297    Levenberg-Marquardt = 10
Free parameter values:
    59.99    0.9572    30.84    0.06554
Total time = 268.2145s    Reduced Chi^2 = 2.1302    Levenberg-Marquardt = 1000
Free parameter values:
    59.92    0.9569    30.85    0.06557
*** No improvement in chi-squared over previous iteration ***
Total time = 268.2218s    Reduced Chi^2 = 2.1282
Free parameter values:
    59.91    0.9569    30.85    0.06558

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	59.91	+/-	2.194
4	0.9569	+/-	0.03198
5	0		
6	30.85	+/-	0.271
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.06558	+/-	0.004789
2	-0.008503		
3	0		

Covariance matrix for free parameters:

1.0000	0.7970	0.6736	-0.9469
0.7970	1.0000	0.7199	-0.9421
0.6736	0.7199	1.0000	-0.7877
-0.9469	-0.9421	-0.7877	1.0000

```

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 1262688 pixels -- processing data... -----> included 1198051 pixels

```

Number of points in input object: 1262688

```

Fraction of object selected: 100.00% (= 1262688 points) ; Access speed: 1.76GB/sec
Fraction of object retained: 94.88% (= 1198051 points)

```

```

**** Total time in cut(sqw) : 0.1sec Including:
Data access time fraction: 60.2%; Transf. time frac.: 39.1%; Resulting pix preparation: 0.0%

```

```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 1262222 pixels -- processing data... -----> retained 1197585 pixels

```

Number of points in input object: 1262222

```

Fraction of object selected: 100.00% (= 1262222 points) ; Access speed: 2.08GB/sec
Fraction of object retained: 94.88% (= 1197585 points)

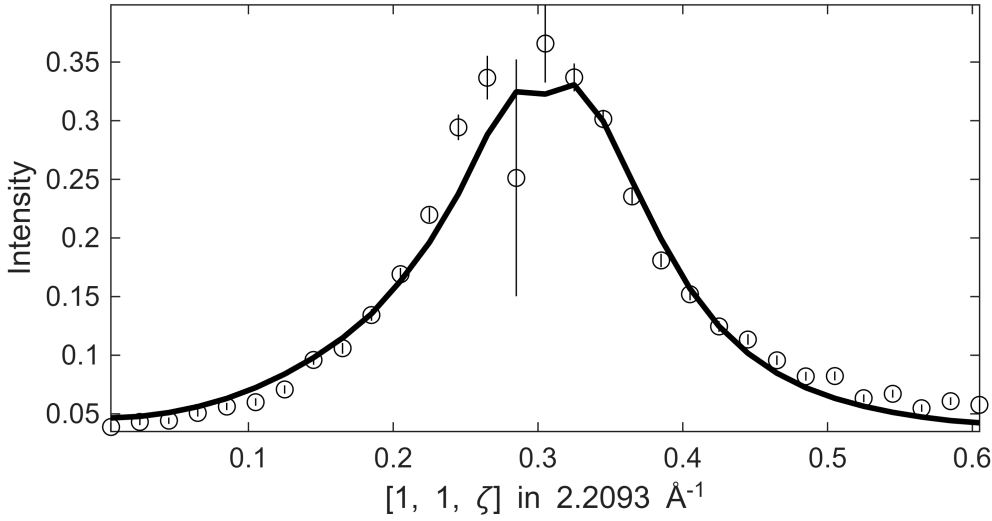
```

```

**** Total time in cut(sqw) : 0.1sec Including:

```

Fe_ei200_align_sym3.sqw
 $S=0.96\pm0.032$; $J0=31\pm0.27$; $\gamma=59.9087\pm2.19419$
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $90 \leq E \leq 1$
 $\zeta=-0.005:0.02:0.615$ in $[1, 1, \zeta]$

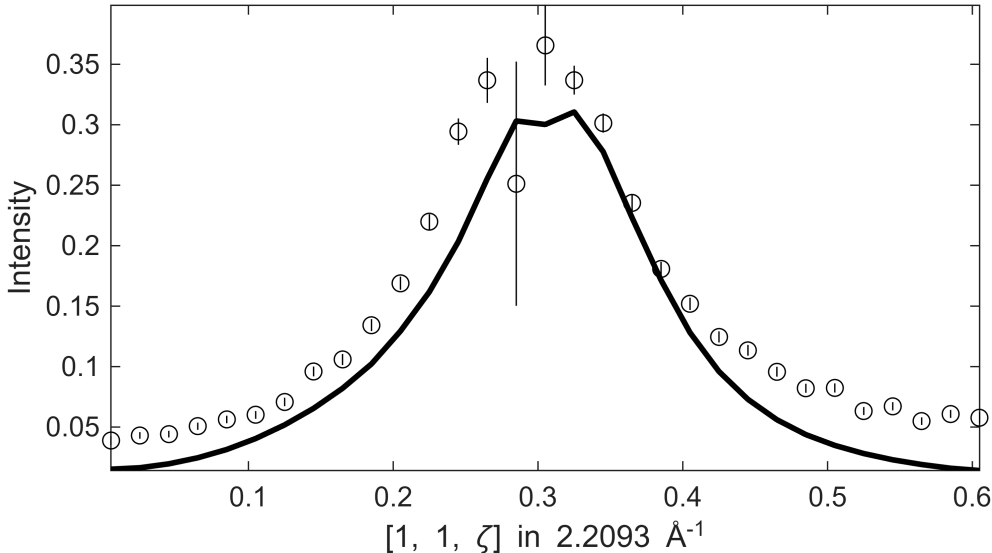


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 1262688 pixels -- processing data... -----> retained 1198051 pixels

Number of points in input object: 1262688
 Fraction of object selected: 100.00% (= 1262688 points) ; Access speed: 2.04GB/sec
 Fraction of object retained: 94.88% (= 1198051 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 29.8%; Transf. time frac.: 49.9%; Resulting pix preparation: 20.3%

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $90 \leq E \leq 1$
 $\zeta=-0.005:0.02:0.615$ in $[1, 1, \zeta]$



 **** En = 110±10

Warning: Dataset:16 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.6921

Beginning fit (max 20 iterations)-----
 Starting point

Total time = 6.7721s Reduced Chi^2 = 56.1009

Free parameter values:

59.91	0.9569	30.85	0.3147
-------	--------	-------	--------

Iteration = 1

Total time = 33.879s Reduced Chi^2 = 1.8547 Levenberg-Marquardt = 0.1

Free parameter values:

51.68	0.7999	30.16	0.08801
-------	--------	-------	---------

Iteration = 2

Total time = 62.6388s Reduced Chi^2 = 1.6897 Levenberg-Marquardt = 0.01

Free parameter values:

57.51	0.9048	30.76	0.06418
-------	--------	-------	---------

Iteration = 3

Total time = 89.6541s Reduced Chi^2 = 1.6854 Levenberg-Marquardt = 0.001

Free parameter values:

59.21	0.9238	30.88	0.06206
-------	--------	-------	---------

Iteration = 4

Total time = 116.9047s Reduced Chi^2 = 1.686 Levenberg-Marquardt = 0.0001

Free parameter values:

59.65	0.9274	30.9	0.06164
-------	--------	------	---------

Total time = 123.7703s Reduced Chi^2 = 1.6849 Levenberg-Marquardt = 0.001

Free parameter values:

59.65	0.9274	30.9	0.06165
-------	--------	------	---------

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	59.65	+/-	4.312
4	0.9274	+/-	0.04631
5	0		
6	30.9	+/-	0.5039
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.06165	+/-	0.006075
2	-0.009175		
3	0		

Covariance matrix for free parameters:

1.0000	0.8678	0.8635	-0.7055
0.8678	1.0000	0.8464	-0.8477
0.8635	0.8464	1.0000	-0.6322
-0.7055	-0.8477	-0.6322	1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 795801 pixels -- processing data... -----> included 724263 pixels

Number of points in input object: 795801

Fraction of object selected: 100.00% (= 795801 points) ; Access speed: 1.61GB/sec
 Fraction of object retained: 91.01% (= 724263 points)

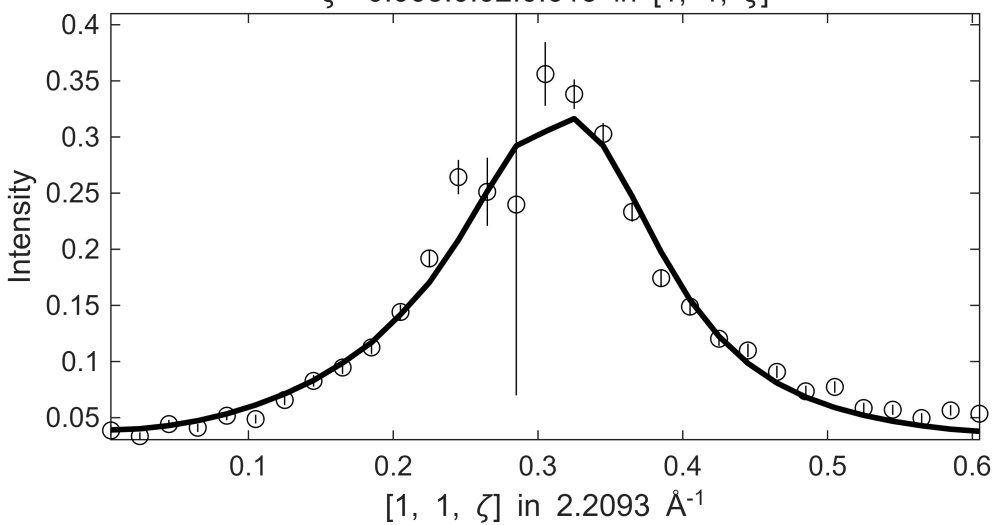
**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 57.5%; Transf. time frac.: 41.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 794224 pixels -- processing data... -----> retained 722741 pixels

Number of points in input object: 794224
 Fraction of object selected: 100.00% (= 794224 points) ; Access speed: 1.67GB/sec
 Fraction of object retained: 91.00% (= 722741 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 30.1%; Transf. time frac.: 52.0%; Resulting pix preparation: 17.8%

Fe_ei200_align_sym3.sqw
 $S=0.93\pm0.046$; $J0=31\pm0.5$; $\gamma=59.6464\pm4.31234$
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $100 \leq E \leq$
 $\zeta=-0.005:0.02:0.615$ in $[1, 1, \zeta]$

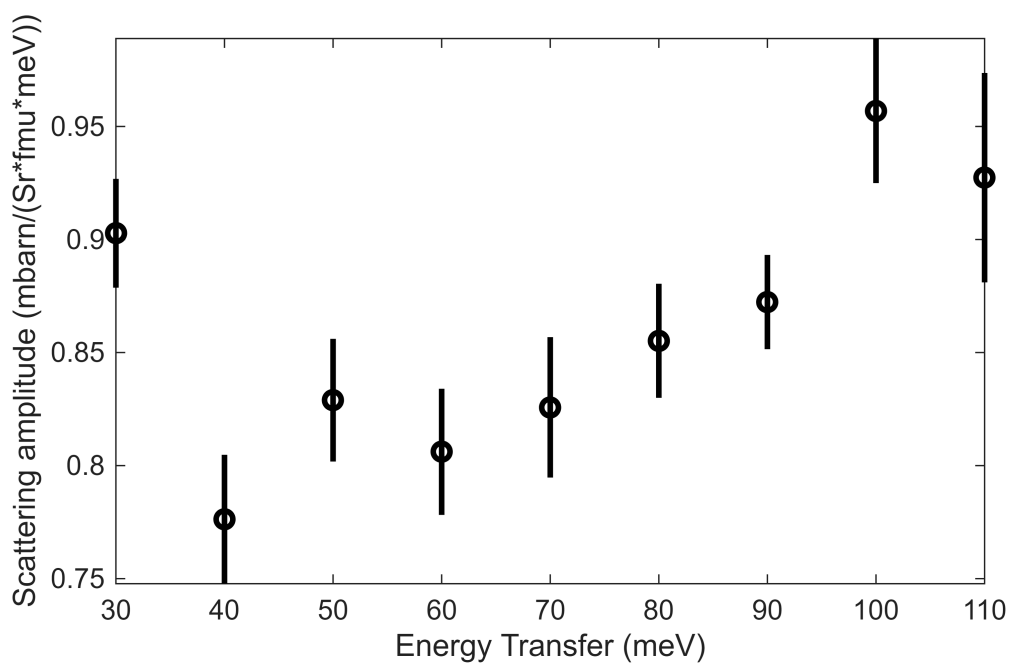
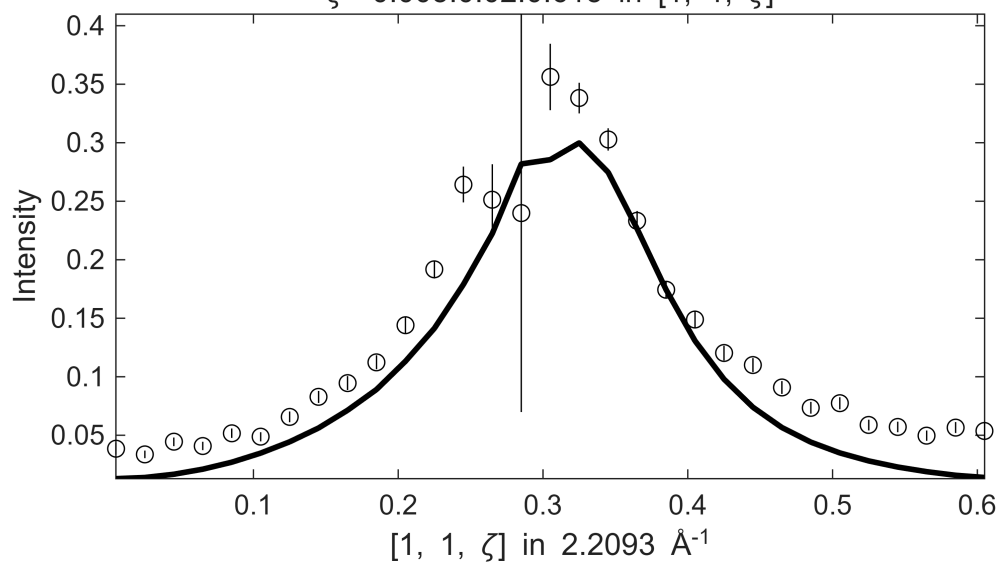


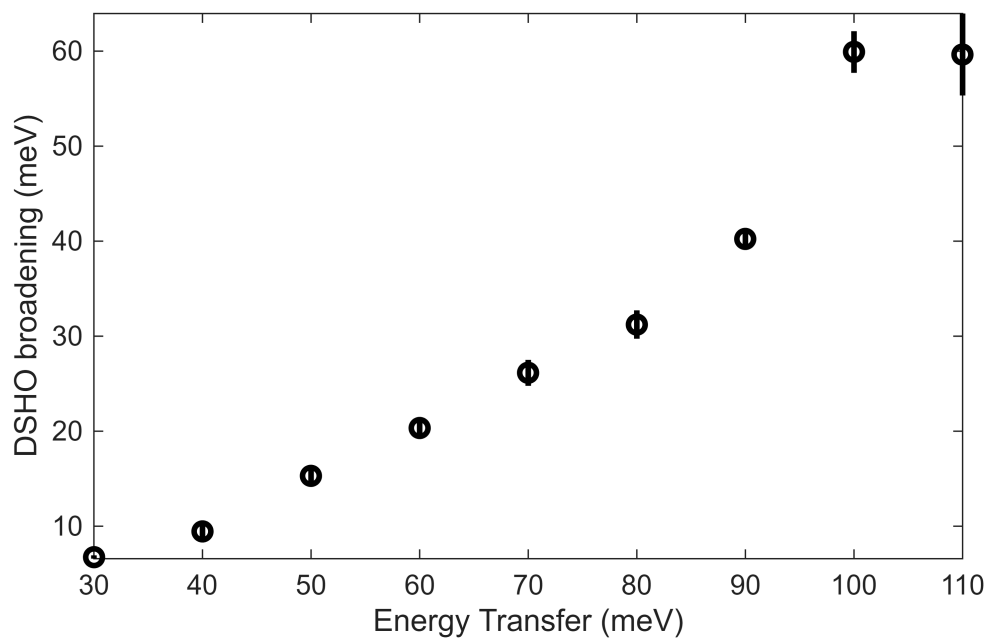
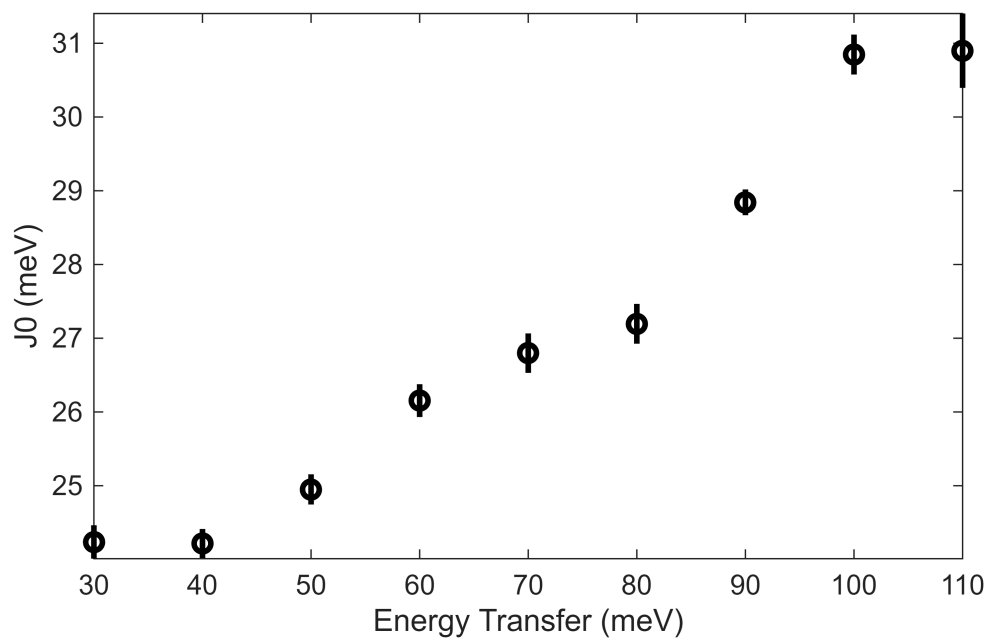
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 795801 pixels -- processing data... -----> retained 724263 pixels

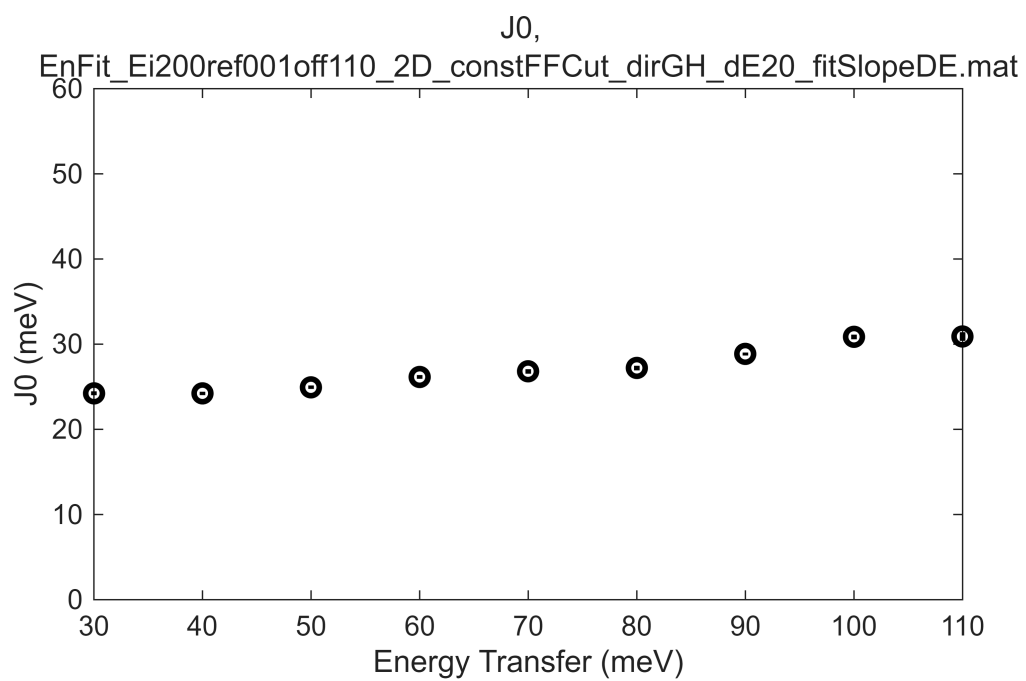
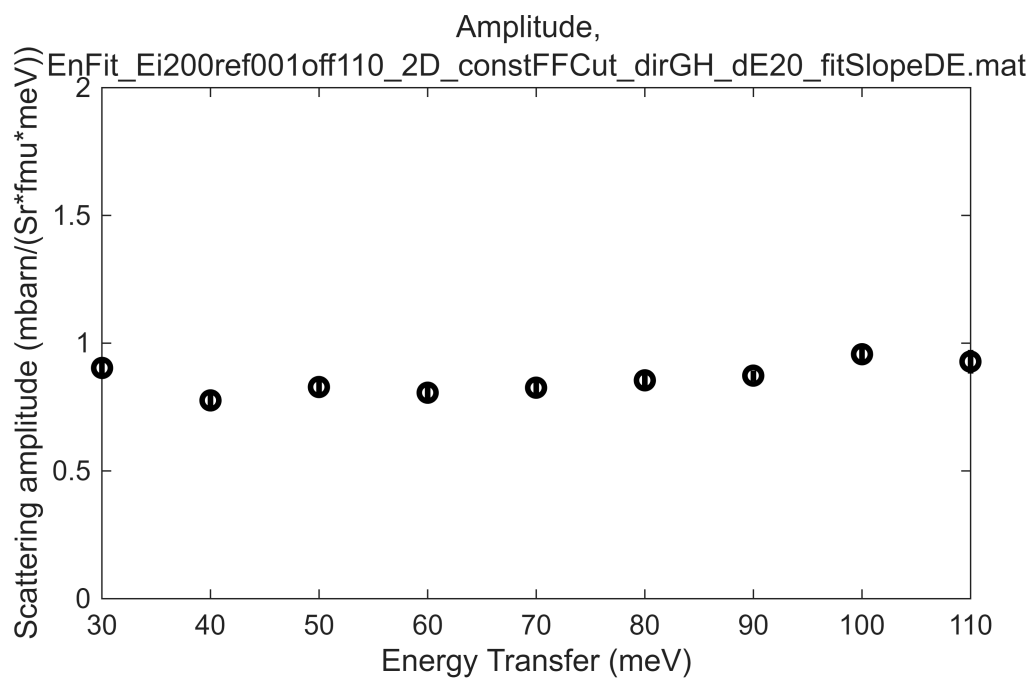
Number of points in input object: 795801
 Fraction of object selected: 100.00% (= 795801 points) ; Access speed: 1.79GB/sec
 Fraction of object retained: 91.01% (= 724263 points)

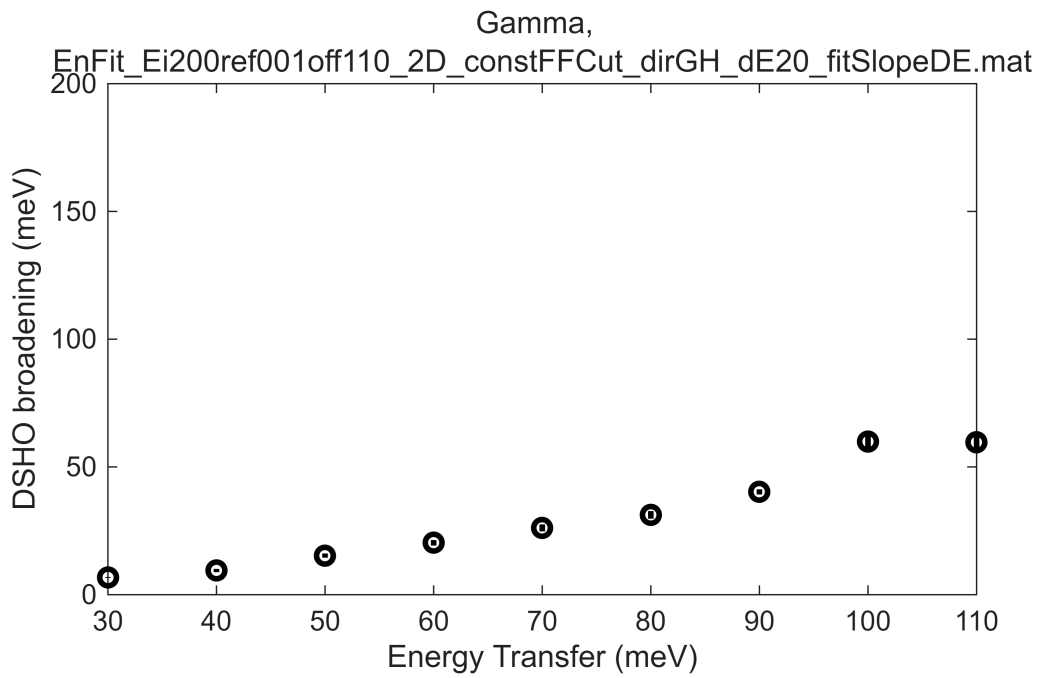
**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 31.8%; Transf. time frac.: 48.3%; Resulting pix preparation: 19.9%

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $100 \leq E \leq 110$ meV
 $\zeta = -0.005:0.02:0.615$ in $[1, 1, \zeta]$



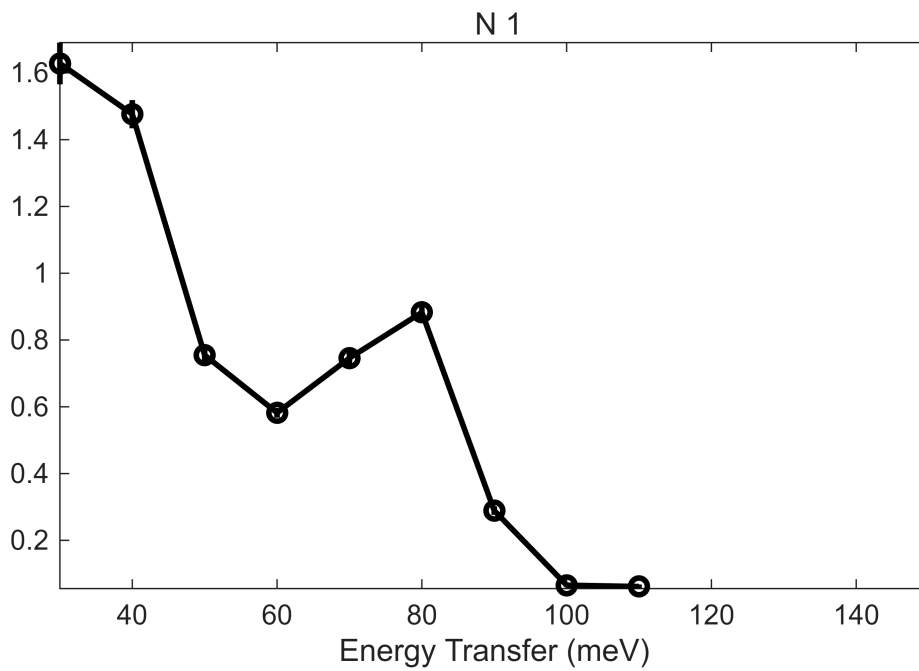




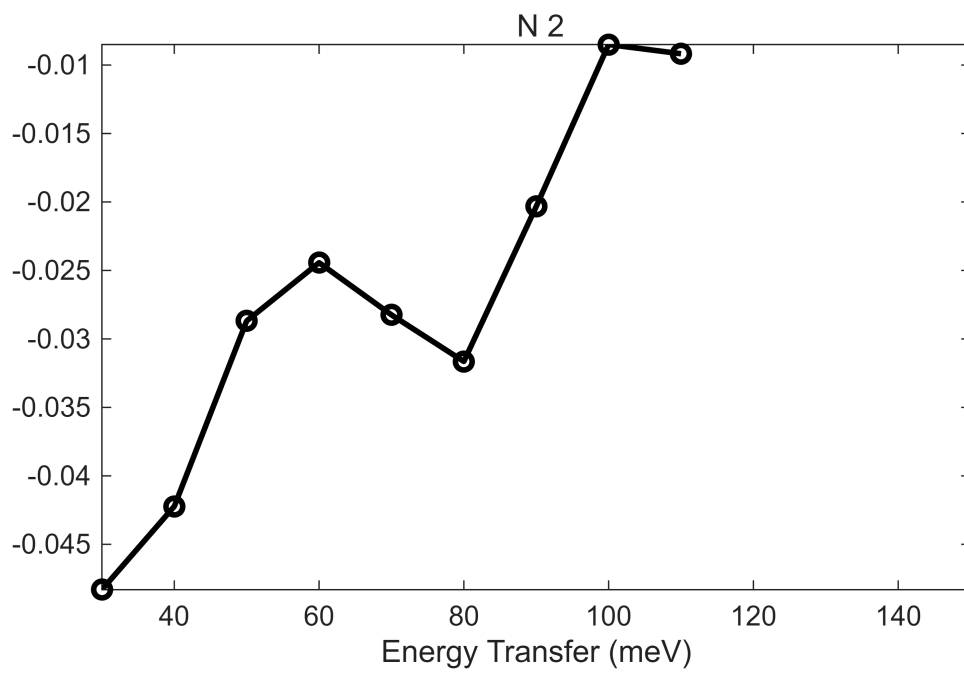


```
%c1,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep
```

```
bg_data = extract_bg_par(fit_res_200fix_bg.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure
```



```
fit_resEi200 = fit_res_200fix_bg;  
save(res_name, "fit_resEi200")
```